Requirements Gathering

MediCare - Stakeholders, User Stories, Use Cases, Functional and Non-functional Requirements

DEPI Full Stack .NET Track | Submission deadline: 16 October 2026

Requirements Gathering

System actors

Patients, doctors, and administrators are the main application actors. Clinic management reviews operational information. A notification service supports appointment messages. The development team and lecturer contribute to delivery and evaluation but are not ordinary application users.

Stakeholder Analysis

Stakeholder	Role / interests	Needs and expectations
Patients	Find doctors and manage appointments and health information.	Search; profiles; times; booking, cancel/reschedule; history; authorized records; reminders.
Doctors	Provide care and coordinate appointments.	Schedule; availability management; relevant patient information; authorized records; updates.
Administrators	Operate platform and maintain accounts/reference data.	Manage users, permissions, doctor profiles, specialties; monitor appointments and statistics.
Clinic management	Oversee clinic operations and service quality.	Appointment trends, schedule coordination, activity insight.
Development team	Build and maintain the project.	Agreed scope, testable requirements, clear technical decisions and feedback.
Lecturer / supervisor	Guide and assess the graduation project.	Evidence of analysis, implementation, testing, documentation and presentation.
Notification service	Deliver appointment-related messages.	Valid contact details, configured events, delivery and error handling.

User Stories

Patient

• As a patient, I want to register and log in securely so that I can use account-based features.

• As a patient, I want to search doctors by specialization, location, and availability so that I can find a suitable doctor.

• As a patient, I want to view a profile and appointment times so that I can choose a consultation.

• As a patient, I want to book an available time so that I can reserve an appointment.

• As a patient, I want confirmation and reminders so that I can keep track of my booking.

• As a patient, I want to cancel or reschedule when policy permits so that I can manage changes.

• As a patient, I want to see appointment history so that I can follow my care.

• As a patient, I want to view authorized records and prescriptions so that I can review treatment information.

• As a patient, I want to upload relevant documents when supported so that I can share visit information.

Doctor

• As a doctor, I want a secure dashboard so that I can manage my work in one place.

• As a doctor, I want to view upcoming and previous appointments so that I can organize my schedule.

• As a doctor, I want to manage availability so that patients can book suitable times.

• As a doctor, I want relevant patient information so that I can prepare for consultations.

• As a doctor, I want to access or manage records according to permissions so that information is handled appropriately.

• As a doctor, I want appointment notifications so that I am aware of changes.

Administrator

• As an administrator, I want to manage patient and doctor accounts so that platform data remains current.

• As an administrator, I want to maintain doctor profiles and specializations so that search results are accurate.

• As an administrator, I want to monitor appointments so that scheduling issues can be addressed.

• As an administrator, I want to manage permissions so that users access only authorized functions.

• As an administrator, I want to view system statistics so that I can monitor activity.

Use Cases

Policies that have not been agreed by the team remain open decisions and are labeled below.

UC-01 Search for a Doctor

Field	Description
Primary actor	Patient
Goal	Find matching doctors.
Preconditions	MediCare is available; guest search depends on project configuration.
Trigger	Patient opens search.
Main flow	1. Enter specialization, location, or availability. 2. System searches and displays matching doctors. 3. Patient opens a profile and views available times.
Alternative flow	No matches: show a message and allow filters to change.
Postconditions / rules	Matching profiles or a no-results message are shown.

UC-02 Book an Appointment

Field	Description
Primary actor	Patient; doctor is supporting actor
Goal	Reserve an available consultation time.
Preconditions	Patient is logged in; doctor has an available slot.
Trigger	Patient selects a slot.
Main flow	1. Open doctor profile. 2. View available times. 3. Select a time and submit. 4. System checks availability and creates appointment. 5. System confirms and sends a notification if configured.
Alternative flow	If the slot was just taken, show updated times and ask the patient to choose again.
Postconditions / rules	Appointment is associated with patient and doctor; overlapping bookings are prevented.

UC-03 Cancel or Reschedule

Field	Description
Primary actor	Patient; doctor is supporting actor
Goal	Change or cancel an existing appointment.
Preconditions	Patient is logged in and owns a manageable appointment.
Trigger	Patient chooses a change action.
Main flow	1. For cancellation, system checks policy, updates status, and confirms. 2. For rescheduling, show times; patient selects one; system validates rules and availability; update and confirm. 3. Notify relevant users if configured.
Alternative flow	If policy disallows the action or the new time is unavailable, explain and offer permitted next steps.
Postconditions / rules	Appointment status/time is updated. Cancellation deadlines and rescheduling rules require confirmation.

UC-04 Manage Availability

Field	Description
Primary actor	Doctor
Goal	Set or update appointment availability.
Preconditions	Doctor is authenticated and authorized.
Trigger	Doctor opens schedule management.
Main flow	1. Select date or period. 2. Add, update, or remove available times. 3. System checks for conflicts with existing appointments. 4. Save and show updated schedule.
Alternative flow	Reject conflicting changes with an explanation.
Postconditions / rules	Availability is updated for patient search and booking.

UC-05 View Appointment Schedule

Field	Description
Primary actor	Doctor
Goal	Review upcoming and previous appointments.
Preconditions	Doctor is logged in.
Trigger	Doctor opens appointment dashboard.
Main flow	1. System loads doctor appointments. 2. Doctor selects a date range or category. 3. System displays appointments and statuses. 4. Doctor opens permitted details.
Alternative flow	If no appointments exist in the range, show an empty state.
Postconditions / rules	Doctor reviews appointments linked to their account.

UC-06 Access Records and Prescriptions

Field	Description
Primary actor	Patient; doctor supports authorized care
Goal	View permitted medical information.
Preconditions	User is authenticated and has permission.
Trigger	Patient opens records or prescriptions.
Main flow	1. System checks identity and permissions. 2. System displays permitted records. 3. Patient selects a record to view.
Alternative flow	No records: show empty state. Unauthorized access: deny access.
Postconditions / rules	Only authorized information is available; exact role permissions need confirmation.

UC-07 Manage Users and Doctor Information

Field	Description
Primary actor	Administrator
Goal	Maintain accounts, doctor information, and permissions.
Preconditions	Administrator is logged in and authorized.
Trigger	Administrator opens management dashboard.
Main flow	1. Select account, doctor profile, or specialty. 2. Edit information or permissions. 3. System validates, saves, and confirms.
Alternative flow	Invalid information produces validation errors and is not saved.
Postconditions / rules	Authorized account/reference data is updated.

UC-08 Monitor Activity and Analytics

Field	Description
Primary actor	Administrator
Goal	Review platform statistics.
Preconditions	Administrator is authorized to view analytics.
Trigger	Administrator opens analytics dashboard.
Main flow	1. System retrieves statistics. 2. Display appointment totals and user activity. 3. Administrator selects a range/category if supported. 4. System refreshes the view.
Alternative flow	If no data is available, show a no-data message.
Postconditions / rules	Administrator reviews available operational metrics.

Functional Requirements

ID	Requirement
FR-01	Authenticate patients, doctors, and administrators.
FR-02	Authorize pages and actions according to assigned user roles.
FR-03	Search doctors by specialization, location, and availability.
FR-04	Display doctor profiles and available appointment times.
FR-05	Allow authenticated patients to book available appointments.
FR-06	Prevent overlapping bookings for the same doctor and time slot.
FR-07	Let patients view appointment history and status in their dedicated Patient Profile dashboard.
FR-08	Allow cancellation/rescheduling under confirmed clinic rules.
FR-09	Let doctors view upcoming and previous appointments.
FR-10	Let authorized doctors manage availability with conflict checks.
FR-11	Allow authorized users to access relevant patient information.
FR-12	Support medical records and prescriptions with role/ownership checks.
FR-13	Support medical document upload (e.g., X-rays, lab results) with secure file storage and extension validation.
FR-14	Generate confirmations and reminders through configured channels.
FR-15	Allow administrators to manage patient/doctor accounts and permissions.
FR-16	Allow administrators to maintain doctor profiles and specializations.
FR-17	Allow administrators to monitor appointments and view agreed statistics.

Non-functional Requirements

ID / attribute	Requirement
NFR-01 Performance	Meet agreed search and booking response targets under a documented test load.
NFR-02 Security	Use secure authentication, role-based authorization, server-side validation, protected secrets, and least privilege.
NFR-03 Privacy	Restrict medical information to authorized users; use synthetic data in demos; define retention/access rules.
NFR-04 Reliability	Preserve data consistency and return a clear result when an appointment slot becomes unavailable.
NFR-05 Usability	Support core tasks on desktop/mobile with clear validation and status feedback.
NFR-06 Accessibility	Use labels, keyboard-operable controls, readable contrast, and clear error messages.
NFR-07 Compatibility	Support agreed current desktop and mobile browsers used for project testing.
NFR-08 Maintainability	Separate presentation, application logic, and data access in the MVC structure.
NFR-09 Availability	Meet an agreed demo/pilot availability target.
NFR-10 Scalability	Keep scheduling/data access extensible utilizing the Repository and Factory design patterns.